

Ahmadreza Farvardin

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EDUCATION

University of Tehran Tehran, Iran
B.Sc. in Mathematics & Computer Science Expected: June 2027
Major GPA: 18.05/20 (Top 5%) | Minor GPA: 17.23/20 | Ranked 2nd out of 40 in Mathematics
Relevant Coursework: Machine Learning, Deep Generative Models, Nonlinear Optimization, Linear Algebra, Statistical Modeling, Mathematical Analysis

RESEARCH INTERESTS

Multimodal Large Language Models, Embodied AI, Generative Models, Computer Vision

TECHNICAL SKILLS

Programming: Python (Advanced), C/C++, MATLAB, Bash
ML/DL Frameworks: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib
Deep Learning: Transformers, CNNs (ResNet, U-Net, YOLO, Faster R-CNN), Attention Mechanisms, LLMs
Specializations: Computer Vision (Segmentation, Detection, Optical Flow), NLP (Machine Translation), Generative Models
Mathematics: Optimization Algorithms, Linear Algebra, Statistical Modeling, Numerical Methods
Tools: Git, Linux, JupyterLab, VS Code, LaTeX

RESEARCH & PROJECTS

- River Velocity Estimation Using Deep Learning & Classical PIV** 🌐 2024
Computer Vision Research Project
- Developed hybrid pipeline combining RAFT optical flow (Princeton) and OpenPIV cross-correlation for automated river surface velocity measurement from video footage
 - Implemented end-to-end system: video preprocessing, RAFT-based flow estimation for unseeded videos, classical PIV for seeded flows, and cross-validation calibration across camera setups
 - Achieved $R^2 = 0.77$, $MAE = 0.36m/s$, $RMSE = 0.58m/s$ on 22 real-world river videos with varying conditions
 - Technologies: PyTorch, RAFT, OpenPIV, OpenCV, NumPy
- Deep Learning Architecture Implementations from Scratch** 2024–2025
Advanced ML Engineering Projects
- Transformer for Machine Translation** 🌐: Implemented complete Transformer (Vaswani et al., 2017) from primitives for German→English translation on Multi30k dataset, including multi-head attention, positional encoding, masking mechanisms, and Noam scheduler
 - U-Net with Attention Gates** 🌐: Built U-Net for binary semantic segmentation on Oxford-IIIT Pets dataset; achieved 0.8375 IoU and 0.9116 Dice score
 - YOLOv3 Object Detection** 🌐: Developed YOLOv3 from scratch on COCO dataset without pretrained weights, implementing anchor boxes, multi-scale predictions, and custom loss functions
 - ResNet-18 Image Classification** 🌐: Implemented ResNet-18 architecture from basic components for CIFAR-10
 - Technologies: PyTorch, NumPy, OpenCV
- Persian Handwritten Mathematical Expression Recognition** 🌐 2024
Machine Learning Course Project — 1st Place, Quera Competition
- Designed end-to-end pipeline: YOLOv8/Faster R-CNN for symbol detection, K-Means/DBSCAN for clustering, CRNN + CTC loss for recognition
 - Achieved 92.3% mAP on detection; implemented semi-supervised learning for data efficiency
 - Technologies: PyTorch, OpenCV, NumPy, Pandas
- Machine Learning & Optimization Algorithms** 2024
Course Projects (ML 🌐 | Optimization 🌐)
- Implemented core ML algorithms from scratch: Naive Bayes, Ridge/Logistic Regression, AdaBoost, PCA, GMM

- Developed optimization algorithms: Gradient Descent, Newton’s Method, Regularized Least Squares with applications to SVMs and polynomial regression
- Technologies: Python, NumPy, Pandas, Scikit-learn, cvxpy, Matplotlib

TEACHING EXPERIENCE

Teaching Assistant	University of Tehran
<i>School of Electrical & Computer Engineering and School of Mathematics, Statistics & CS</i>	
• Nonlinear Optimization — Supervising Professor: Prof. Mohammad Saeed Sarafraz	Fall 2024
• Linear Algebra — Supervising Professor: Prof. Mohammad Javad Yazdanpanah	Spring 2024
• Mathematical Analysis — Supervising Professor: Prof. Fatemah Ayatollah Zadeh Shirazi	Spring 2024
• Calculus I & II — Supervising Professor: Prof. Shaghayegh Mokarami	Fall 2023 – Spring 2024
• Conducted weekly problem-solving recitations, designed assignment solutions, and held regular office hours for undergraduate engineering and mathematics students.	

HONORS & AWARDS

- **1st Place** — Quera Competition for Machine Learning Final Project (Mathematical Expression Recognition), 2024
- **Ranked 2nd out of 40** — Mathematics Major, University of Tehran

ADDITIONAL INFORMATION

Languages: Persian (Native), English (Professional Working Proficiency)
